

Industry Analysis Report

Multi-AI Agent Systems in Global Financial Forecasting

Visual Pipeline Diagrams • System Architecture • Data Flow

11 Companies • USA, United Kingdom, Spain, Canada

B.Tech Capstone Project: Financial Forecasting by Multi-AI Agent System

Prepared for Professor Review

Regions Covered: USA | United Kingdom | Spain | Canada

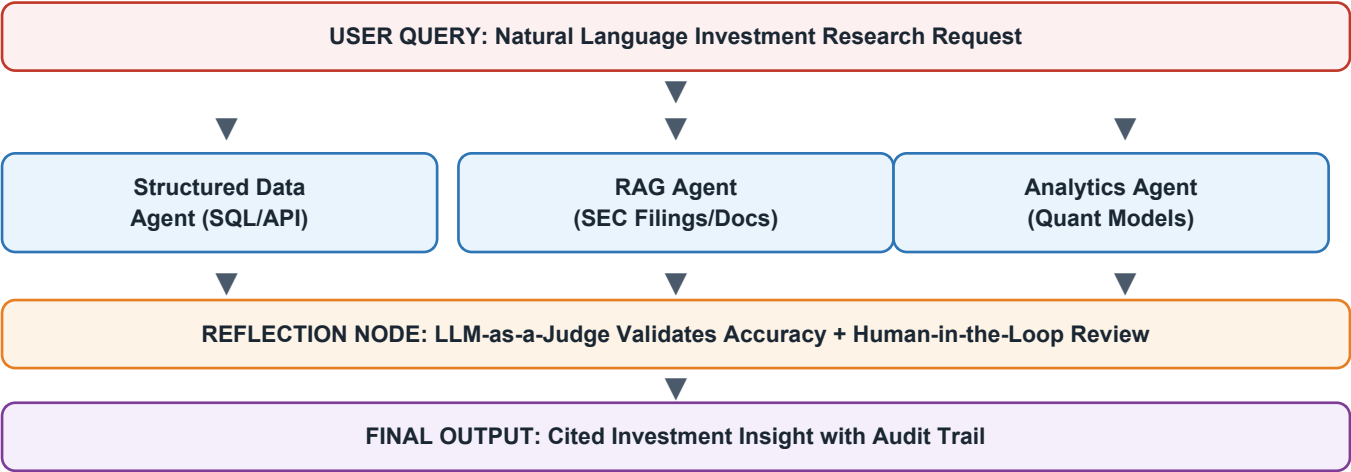
All company facts independently verified, July 2026

JPMorgan Chase

HQ: New York, USA	Sector: Tier-1 Investment Bank	Product: Ask D.A.V.I.D. & IndexGPT	Tech: LangGraph, GPT-4, FactSet, SEC EDGAR
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JPMorgan Chase, the world's largest bank by market capitalisation, built **Ask D.A.V.I.D.** (Data Analytics, Visualization, Insights and Decision-making assistant) - a production-grade multi-agent AI system for its Private Bank investment research team. It is built on **LangGraph**, the exact agentic framework recommended for this BTP project, and reports over 90% accuracy on domain-specific queries.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. User submits a natural-language investment query 2. Supervisor Agent parses intent and decomposes it into sub-tasks 3. Supervisor routes to Structured Data Agent, RAG Agent, or Analytics Agent 4. Worker agents execute in parallel and return results to shared state 5. Supervisor may re-route for follow-up analysis (iterative loop) 6. Reflection Node validates accuracy and completeness (LLM-as-a-judge) 7. Human reviewer approves or requests revision 8. Final investment insight delivered with source citations

Key Lesson for Your BTP: JPMorgan's architecture is the closest real-world blueprint for this BTP. Their Supervisor-Worker pattern maps directly onto the Orchestrator Agent; their Reflection Node is the equivalent of the Chief Risk Officer Agent that validates all outputs before the final forecast. Use LangGraph's StateGraph with conditional edges for routing.

Morgan Stanley

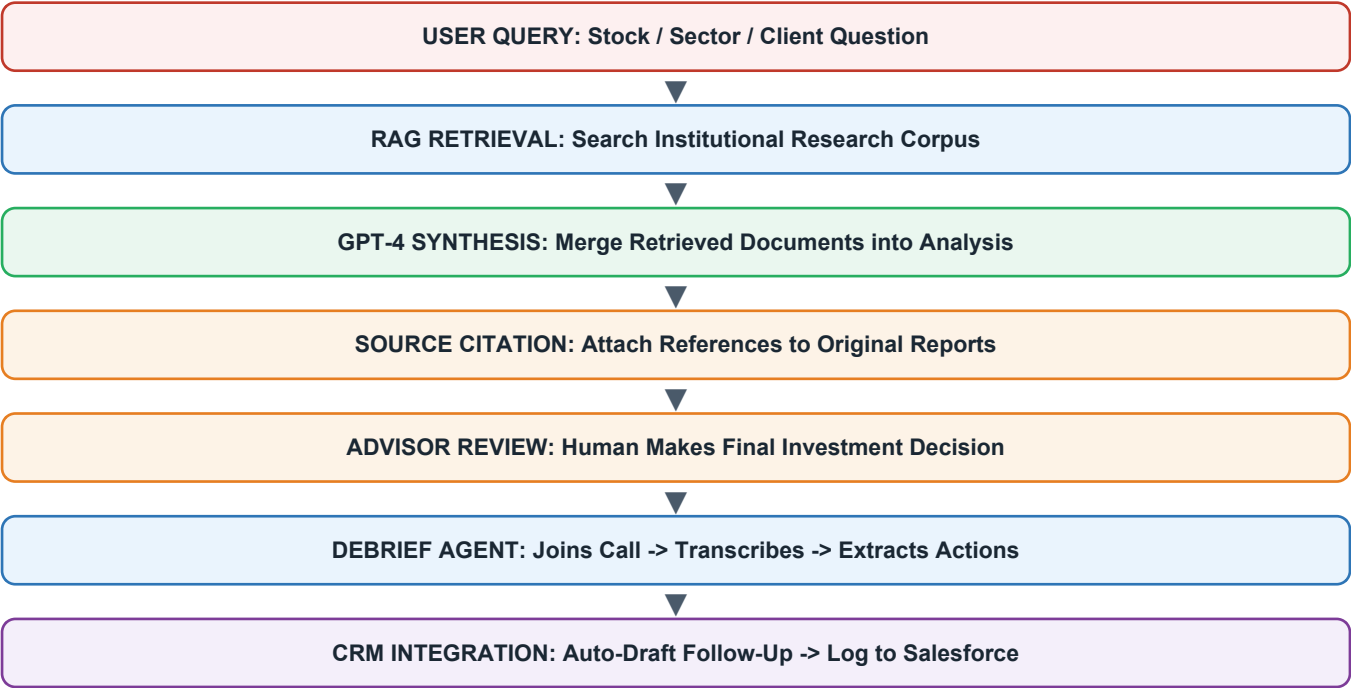
HQ: New York, USA	Sector: Tier-1 Investment Bank	Product: AskResearchGPT & AI @ Morgan Stanley Debrief	Tech: OpenAI GPT-4, Salesforce, Zoom API
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Morgan Stanley partnered with **OpenAI** to build AI systems across the firm. **Note that these are two distinct products serving two different divisions**, a distinction frequently blurred in secondary reporting.

AskResearchGPT (launched October 2024) serves the **Investment Banking, Sales & Trading and Research** divisions, synthesising the firm's roughly **70,000 proprietary research reports published annually**.

AI @ Morgan Stanley Debrief serves **Wealth Management**, whose roughly 15,000 financial advisors reached **98% adoption** within months of the June 2024 firm-wide rollout.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. User queries the system about a stock, sector or client situation
2. RAG retrieval searches the institutional research corpus
3. GPT-4 synthesises retrieved documents into investment analysis
4. Response delivered with source citations back to original reports
5. (Debrief) AI agent joins the client meeting with explicit consent
6. Real-time transcription and extraction of client needs
7. Auto-generates meeting notes, action items and a follow-up email draft
8. Advisor reviews, edits and sends - logged to Salesforce CRM

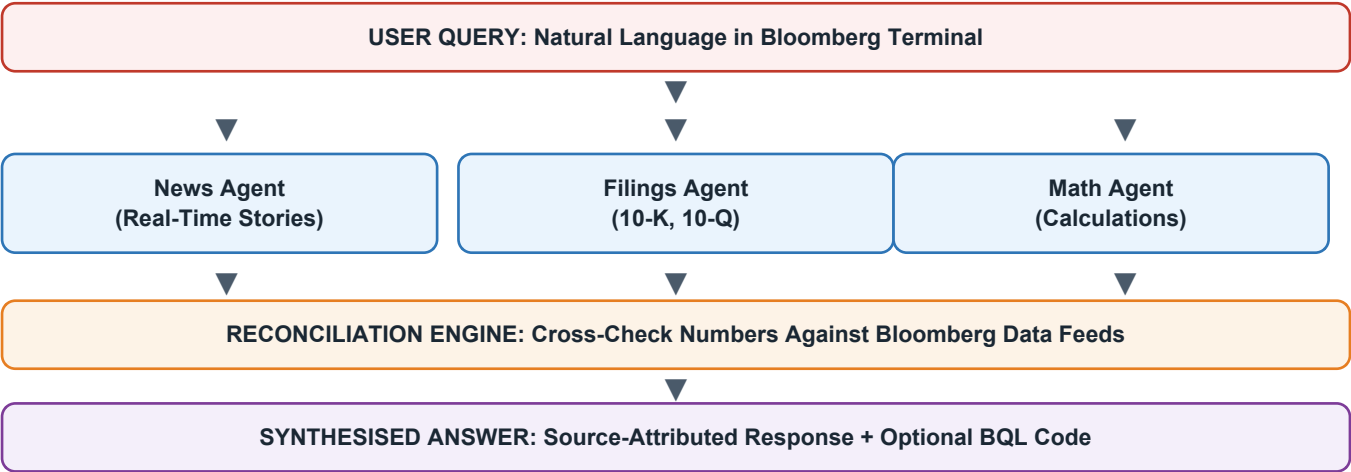
Key Lesson for Your BTP: Morgan Stanley demonstrates the 'human-in-the-loop' synthesis model: the AI does the heavy research, the human makes the final call. That is exactly the role of the Chief Risk Officer Agent in this project - it synthesises the agent debate, but the output is a recommendation, not an execution.

Bloomberg L.P.

HQ: New York, USA	Sector: Financial Data & Terminal Giant	Product: BloombergGPT & ASKB Agentic AI	Tech: Multi-LLM over MCP, FinPile, BQL
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Bloomberg trained **BloombergGPT**, a custom 50-billion-parameter financial LLM on decades of proprietary financial data ('FinPile'). It then evolved that capability into **ASKB**, a multi-agent agentic AI system embedded inside the Bloomberg Terminal, launched in **February 2026** and currently in beta.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. User submits a natural-language query in the Bloomberg Terminal
2. Controller Agent analyses intent and breaks it into sub-tasks
3. News Agent, Filings Agent and Math Agent are dispatched IN PARALLEL
4. News Agent retrieves real-time stories and sentiment scores
5. Filings Agent extracts data from SEC filings and earnings transcripts
6. Math Agent performs quantitative calculations on structured data
7. Reconciliation Engine validates numbers against Bloomberg feeds (mismatches raise warnings)
8. Controller merges outputs, resolves conflicts, formats with source attribution
9. Final synthesised answer delivered, with optional BQL code for quants

Key Lesson for Your BTP: Bloomberg's Controller Agent pattern is the blueprint for the Orchestrator. More importantly, their **reconciliation engine** - cross-checking every AI-generated number against real data - is the single most important feature to copy. Validate agent forecasts against actual market data so the system cannot hallucinate a price.

Man Group

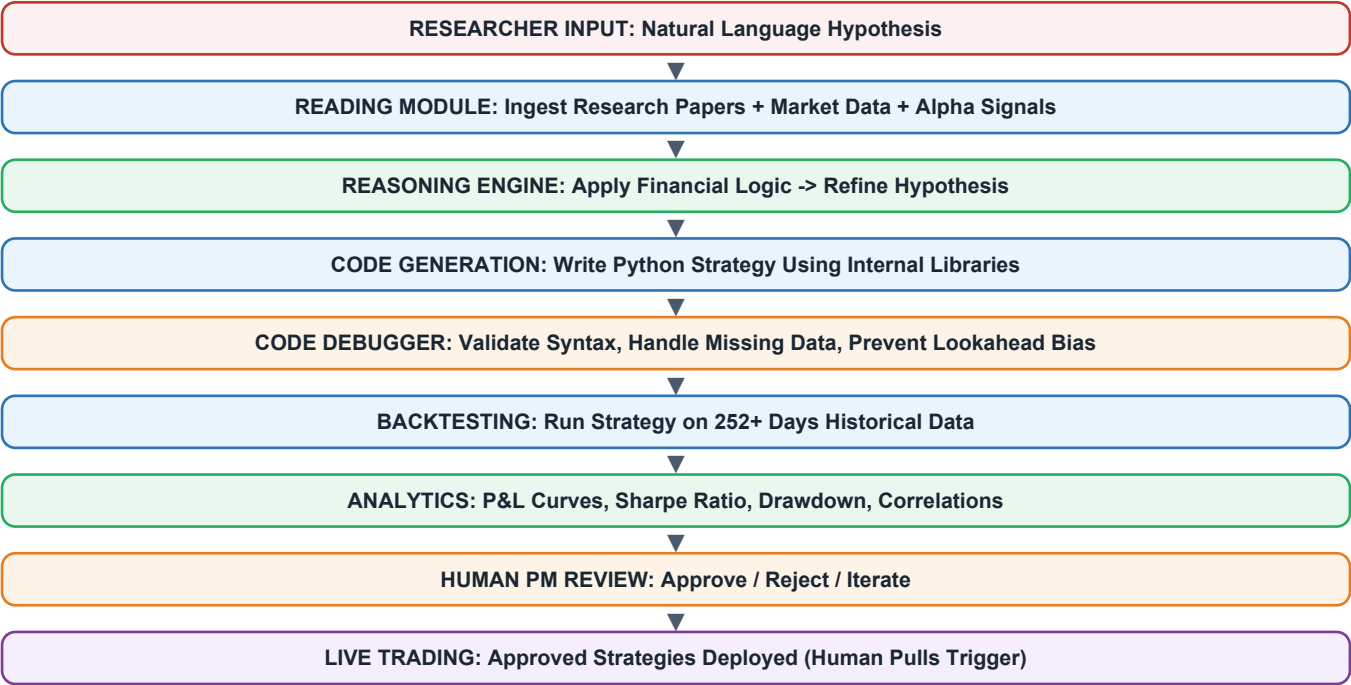
HQ: London, UK	Sector: World's Largest Listed Hedge Fund	Product: AlphaGPT & Alpha Assistant	Tech: Proprietary LLMs, Python, quant libraries
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Man Group manages **\$227.6 billion** in assets (31 December 2025), rising to **\$228.7 billion** at 31 March 2026. **These are two separate tools**, often incorrectly reported as one:

AlphaGPT, built by quant arm Man Numeric, is an agentic system that autonomously generates, codes and backtests trading strategies, mimicking the research process of a human quant analyst. Several dozen AlphaGPT-generated signals have already been approved for live trading.

Alpha Assistant is an AI-powered coding agent that helps researchers run proprietary quantitative analysis using internal libraries and data through natural-language instructions.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. Researcher describes an investment hypothesis in natural language 2. Orchestrator agent interprets the objective and selects tools 3. Reading module ingests relevant research papers and market data 4. Reasoning engine processes data and refines the hypothesis 5. Code generation agent writes Python strategy code using internal libraries 6. Code debugger validates syntax and handles edge cases (missing data, lookahead bias) 7. Backtesting agent runs the strategy on historical data (252+ market days) 8. Performance report generated: account curves, Sharpe ratio, drawdown, correlations 9. Human PM reviews results - approves, rejects, or sends back for iteration 10. Approved strategies go to live trading (a human pulls the trigger)

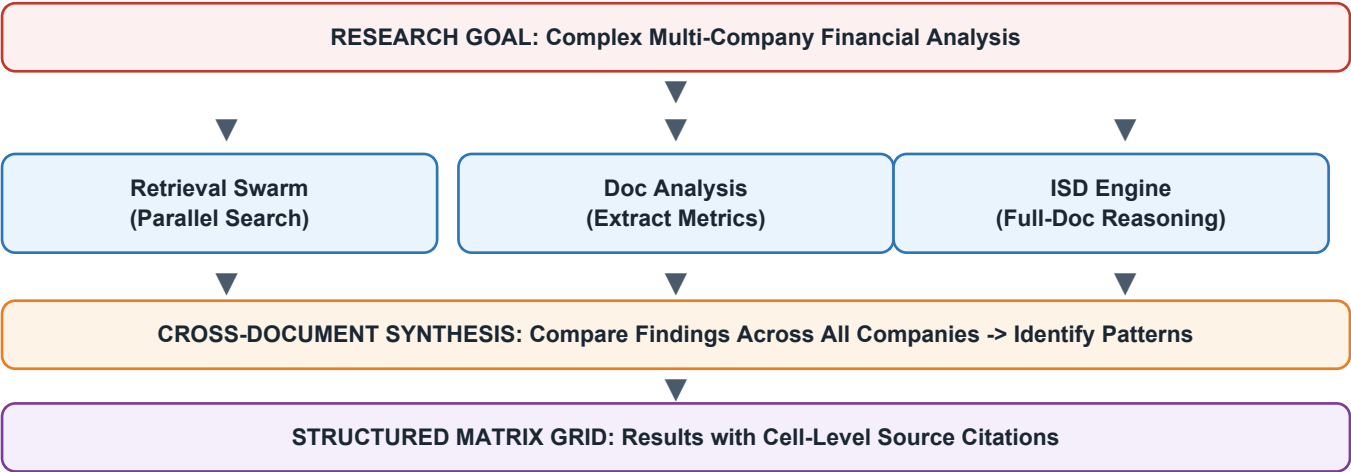
Key Lesson for Your BTP: Man Group's 'hypothesis -> code -> backtest -> human review' loop maps directly onto Phase 5 of this project. The system should generate **backtestable** forecasts that can be validated against real historical price movements - not just narrative predictions.

Hebbia

HQ: New York, USA	Sector: AI Research Platform for Wall Street	Product: Hebbia Matrix	Tech: Multi-agent engine, model-agnostic, ISD
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Hebbia built **Matrix** - a multi-agent orchestration engine used by top-tier Wall Street investment banks and private equity firms. It dispatches autonomous AI agents in parallel across thousands of financial documents (SEC 10-K/10-Q filings, investor decks, spreadsheets) to perform deep due-diligence research that would take human analysts weeks.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. User defines a complex research question (e.g. 'Compare revenue trends of 50 SaaS companies') 2. Planning Agent decomposes it into sub-tasks per company / document set 3. Retrieval Agent Swarm dispatched IN PARALLEL across all document sets 4. ISD breaks each document into granular chunks (beyond simple RAG) 5. Document Analysis Agents extract specific metrics from each filing 6. Cross-Document Synthesis Agent compares findings across all companies 7. Patterns, outliers and contradictions identified 8. Results formatted into a structured Matrix grid with full source citations 9. User can drill down into any cell to see the underlying source documents

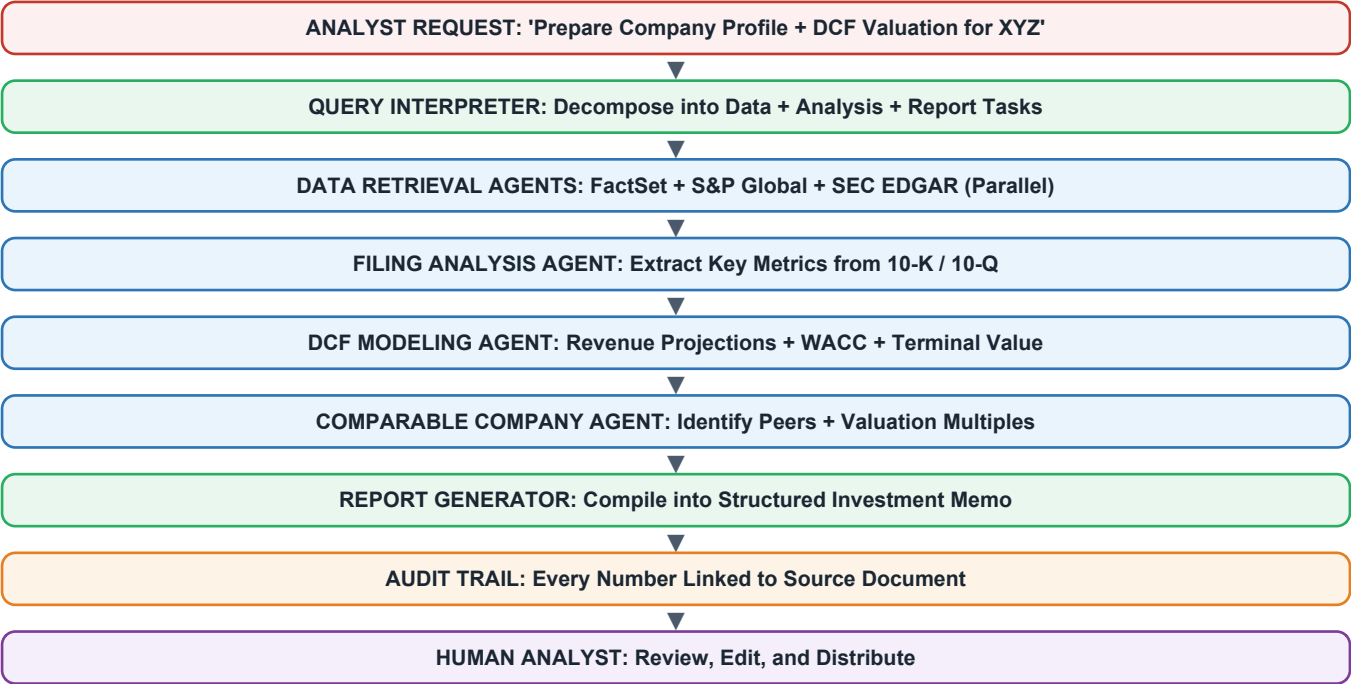
Key Lesson for Your BTP: Hebbia's parallel agent dispatch is what the Orchestrator should do - send the Technical, Fundamental, Sentiment and Macro agents to work simultaneously, then merge their outputs into a unified analysis before the debate phase begins.

Rogo

HQ: New York, USA	Sector: Enterprise AI for Investment Banking	Product: AI Analyst Swarm Platform	Tech: FactSet, S&P Global, SEC EDGAR, Pydantic
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Rogo provides an enterprise AI platform built specifically for investment banks, hedge funds and private equity firms. Founded in 2022 by Gabriel Stengel, John Willett and Tumas Rackaitis, it functions as an **'AI Analyst Swarm'**: autonomous agents pull data from FactSet and S&P Global, parse SEC filings, build DCF valuation models and auto-generate professional investment memos. Over 35,000 financial professionals at 250+ institutions use it. Rogo raised a **\$160M Series D** led by Kleiner Perkins in April 2026 at a **\$2B valuation** (\$310M raised in total).

Pipeline Architecture Diagram



Pipeline Flow Summary

1. Analyst submits a research request (e.g. 'Prepare a company profile for XYZ with DCF valuation') 2. Query Interpreter breaks the request into data retrieval + analysis + report tasks 3. Data Retrieval Agents fetch financials from FactSet, S&P Global and SEC EDGAR in parallel 4. Filing Analysis Agent extracts key metrics from 10-K / 10-Q filings 5. Financial Modeling Agent builds a DCF model with revenue projections and WACC 6. Comparable Company Agent identifies the peer group and valuation multiples 7. Report Generation Agent compiles everything into a structured investment memo 8. Audit trail attached - every number linked to its source document 9. Human analyst reviews, edits and distributes

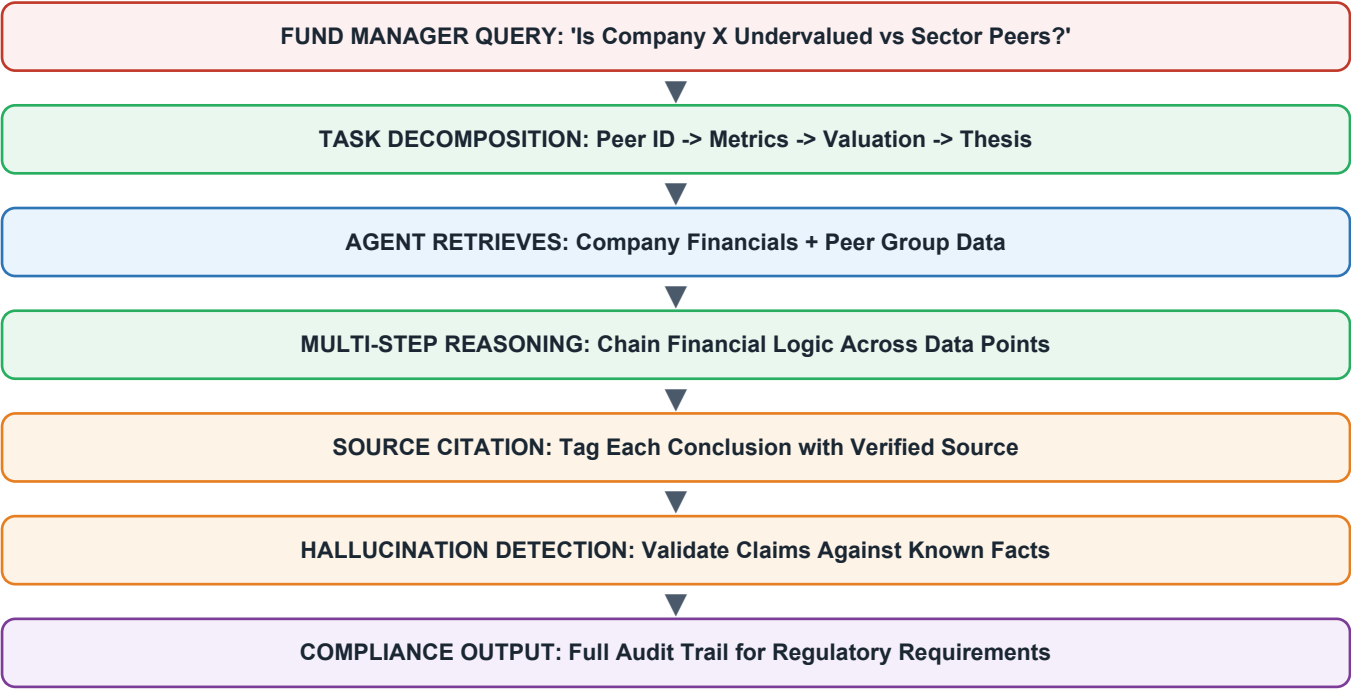
Key Lesson for Your BTP: Rogo's 'data retrieval -> analysis -> report generation' pipeline is this project's exact flow: fetch market data, run specialist agents, synthesise a forecast report. Implement their audit trail so every forecast number is traceable to yfinance or a news source.

Samaya AI

HQ: Mountain View, California, USA	Sector: Institutional Hedge Fund AI Platform	Product: Expert AI Agents for Financial Reasoning	Tech: Proprietary reasoning LLM, citation engine
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Samaya AI was founded in 2022 by **Maithra Raghu** (formerly Google Brain) and **Fabio Petroni**, and is headquartered in **Mountain View, California**. It deploys **expert AI agents** for leading hedge funds and private equity firms. Its agents perform multi-step financial reasoning, investment research, deal diligence and client advisory - with full audit trails and source citations explicitly designed to combat LLM hallucination.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. Fund manager submits a complex question (e.g. 'Is Company X undervalued vs sector peers?') 2. Task Decomposition breaks it into: peer identification -> metric extraction -> valuation comparison -> thesis 3. Agent retrieves company financials and peer group data 4. Multi-step reasoning chains financial logic across data points 5. Each intermediate conclusion is tagged with a source citation 6. Hallucination detection validates claims against verified data 7. Final analysis compiled with a full audit trail 8. Compliance-ready output delivered to the fund manager for review

Key Lesson for Your BTP: Samaya's emphasis on source citation and anti-hallucination is critical for this project. When the Technical Agent says 'RSI is oversold at 28', it should cite the exact yfinance data point. When the News Agent says 'negative sentiment detected', it should link to the actual article.

RavenPack / Bigdata.com

HQ: Marbella, Spain / New York, USA	Sector: Financial Data & Sentiment Intelligence	Product: Bigdata.com Agentic AI Platform	Tech: NLP, deep learning, 40,000+ sources
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RavenPack, a long-established leader in financial analytics data, launched **Bigdata.com** - an agentic AI platform that grounds autonomous research agents in trusted, high-quality financial data. It processes millions of documents daily from **40,000+ sources in 13 languages**, and its Briefs product analyses over 55,000 global news sources. Financial institutions deploy these agents for 24/7 investment monitoring and real-time market event analysis.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. Content Collection ingests millions of articles and filings daily from 40,000+ sources 2. Text Extraction normalises raw documents into machine-readable format 3. NER Agent identifies companies, people, currencies and financial entities 4. Event Detection Agent classifies text into an event taxonomy (merger, earnings, legal, etc.) 5. Sentiment Agent calculates sentiment, relevance and impact scores per entity 6. Relevance and novelty scoring filters signal from noise 7. Structured data delivered via real-time API and Snowflake integration 8. Autonomous Research Agents use the enriched data for 24/7 portfolio monitoring 9. Alerts triggered when sentiment or events cross user-defined thresholds

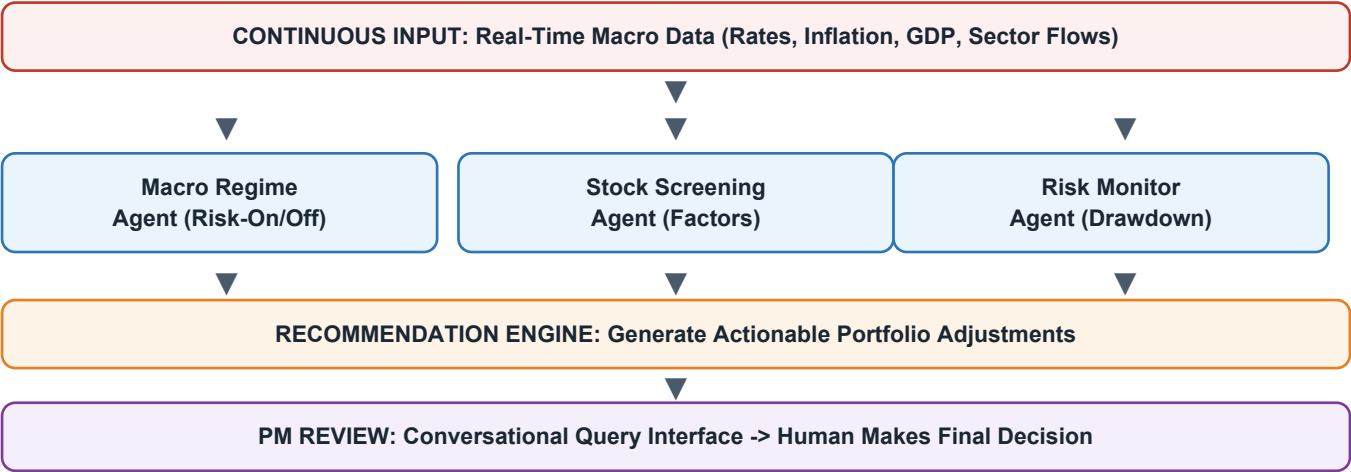
Key Lesson for Your BTP: RavenPack's sentiment methodology is the blueprint for the News & Sentiment Agent. Two things to copy: score **relevance and impact alongside polarity** (not just positive/negative), and ground agents in verified data feeds so they fetch real market data rather than hallucinating numbers.

Boosted.ai

HQ: Toronto, Canada / New York, USA	Sector: Institutional Portfolio AI	Product: Agentic Portfolio Management Platform & Alfa	Tech: ML models, factor engine, real-time feeds
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Boosted.ai provides AI-powered investment research tools that automate portfolio management tasks for institutional asset managers. The company has raised **\$71M** and employs roughly 75 people. Its platform deploys specialised research agents that continuously monitor portfolio risks, detect macro regime shifts and generate stock and sector ideas. Its newer tool, **Alfa**, is described as an AI coworker that users can train to think like them.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. System continuously ingests macro data (rates, inflation, GDP, sector flows)
2. Macro Regime Agent classifies the current environment (e.g. 'risk-off', 'inflationary')
3. Stock Screening Agent filters the equity universe based on regime + factor criteria
4. Risk Agent evaluates current portfolio exposure and concentration
5. System generates actionable recommendations (add / reduce positions)
6. PM queries the system via natural language for deeper analysis
7. Conversational Agent retrieves and synthesises relevant data
8. PM reviews recommendations and makes the final decision

Key Lesson for Your BTP: Boosted.ai's Macro Regime Detection Agent maps directly onto the Macroeconomic Agent, and their Risk Monitoring Agent onto the Chief Risk Officer. The key insight: agents should not just forecast - they should assess the **risk and confidence level of their own predictions**.

Danelfin

HQ: Barcelona, Spain

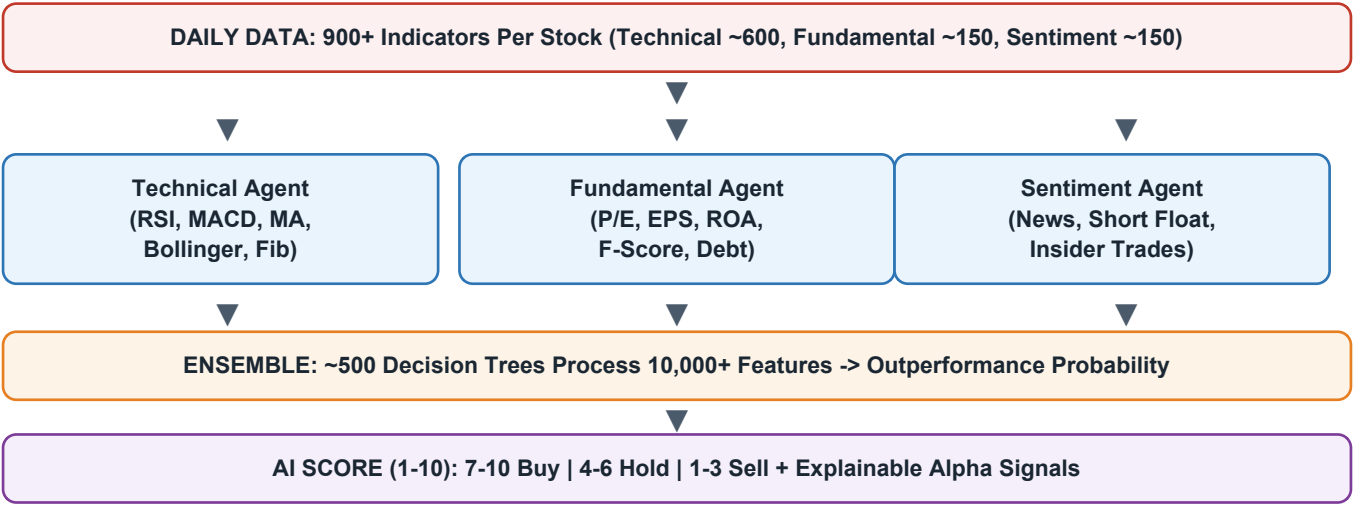
Sector: AI Equity Rating Platform

Product: AI Score System (Technical + Fundamental + Sentiment Agents)

Tech: ~500 decision trees, 900+ indicators, XAI

Danelfin, founded in 2018 by CEO Tomas Diago and headquartered in **Barcelona**, is the most directly comparable commercial product to this BTP project. It combines **10,000+ daily features** via parallel Technical, Fundamental and Sentiment agents to compute an overall '**AI Score**' (1-10) rating each stock's probability of outperforming the market over the next 90 days. It covers all US-listed stocks and ETFs, and as of February 2026 expanded European coverage to 5,500+ stocks.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. Daily collection of 900+ indicators per stock (Technical ~600, Fundamental ~150, Sentiment ~150) 2. Feature engineering transforms 900 indicators into 10,000+ features (cross-analysis, peer decile ranking) 3. Rolling window training on the last 252 market days (US) / 500 days (Europe) 4. ML identifies 'alpha signals' - features that historically preceded outperformance 5. Technical Agent scores technical alpha signals -> Technical subscore (1-10) 6. Fundamental Agent scores fundamental alpha signals -> Fundamental subscore (1-10) 7. Sentiment Agent scores sentiment alpha signals -> Sentiment subscore (1-10) 8. 500 decision trees ensemble ALL features into an outperformance probability 9. Probability mapped to a final AI Score (1-10): 7-10 = Buy, 4-6 = Hold, 1-3 = Sell 10. Explainable AI surfaces the top positive and negative alpha signals driving the score

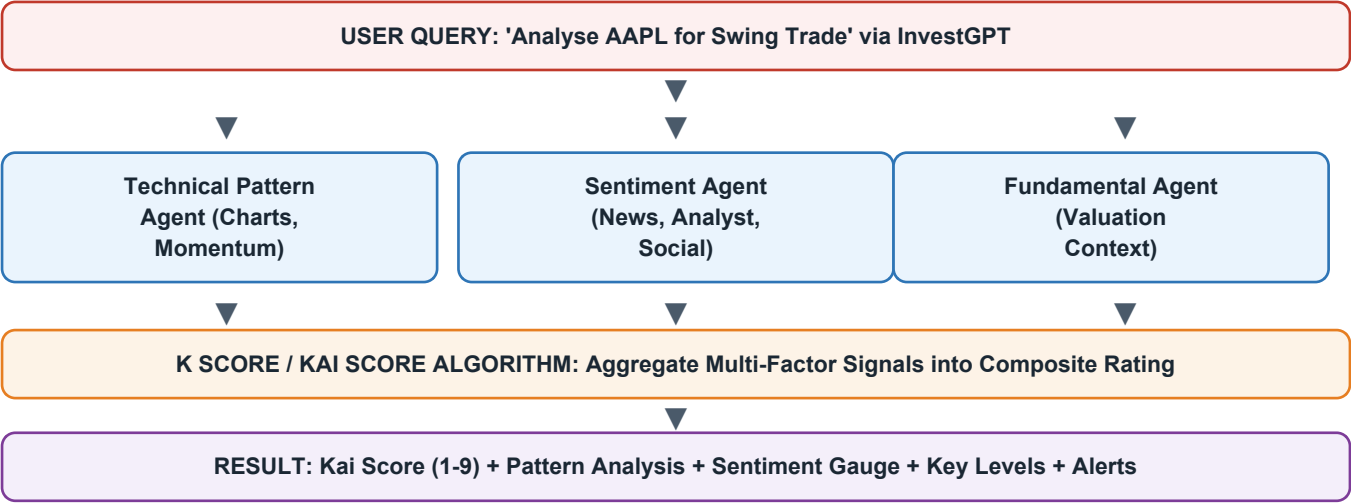
Key Lesson for Your BTP: Danelfin is the closest commercial benchmark to this project. Their 3-pillar approach (Technical + Fundamental + Sentiment) is identical to the specialist agent structure here. Study danelfin.com/how-it-works to understand how they weight and combine agent outputs into a single score. **This project's differentiator is the Bull vs Bear adversarial debate layer, which Danelfin does not have.**

Kavout

HQ: Seattle, Washington, USA	Sector: AI Stock Intelligence Platform	Product: InvestGPT & K Score / Kai Score	Tech: K Score / Kai Score, NLP sentiment, patterns
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Kavout provides an agentic AI investing platform featuring **InvestGPT** (a financial research copilot) and two related equity ratings. **K Score** is the long-standing institutional rating, an AI model built on deep learning and alternative data used by hedge funds. **Kai Score** (scaled **1-9**) is a retail-facing version of the same engine, built for Kavout's online product in response to demand from individual investors - **the two coexist; it is not a rename**. Both evaluate equities using parallel sentiment, momentum and technical pattern agents across 9,000+ US stocks and 20+ global markets.

Pipeline Architecture Diagram



Pipeline Flow Summary

1. User queries InvestGPT with natural language (e.g. 'Analyse AAPL for a swing trade')
2. InvestGPT orchestrates data retrieval from market data APIs
3. Technical Pattern Agent scans multiple timeframes for chart patterns and momentum
4. Sentiment Agent evaluates news flow, analyst consensus and social signals
5. Fundamental data pulled for valuation context
6. Kai Score algorithm aggregates all factor signals into a composite rating
7. Results presented: Kai Score, pattern analysis, sentiment gauge, key levels
8. Real-time alerts set for significant score changes or pattern breaks

Key Lesson for Your BTP: Kavout's Kai Score concept - combining multiple agent scores into one final rating - is exactly what the Chief Risk Officer Agent should produce: a single composite forecast score with a confidence percentage, derived from all specialist agent inputs.